

SHIFT ROSTER & CLAIMS MANAGEMENT SYSTEM

Project Name: Shift Roster & Claims Management System

Prepared For: Development Team

1. PROJECT SUMMARY

1.1. Purpose of This System

The Shift Roster & Claims Management System will automate employee scheduling, claims submission, payroll calculations, and compliance monitoring.

This system will replace manual processes with a centralized web-based solution.

1.2. Main Business Problem

The organization currently faces:

- Manual shift scheduling errors
- Incorrect payroll calculations
- Lack of compliance validation
- Slow claim approval processes
- Difficulty tracking employee work hours

2. PRIMARY SYSTEM OBJECTIVES

Developers must build a system that:

1. Manages employee records
2. Generates rotation-based shift rosters
3. Assigns employees to shifts
4. Allows employees to submit claims
5. Allows admins to approve claims
6. Calculates payroll automatically
7. Validates labour compliance rules
8. Generates reports

3. USER ROLES

Only **two roles exist** in the system.

3.1 Admin Role

Admin users have **full access**.

Admin Responsibilities:

- Create employees
- Create teams
- Create shifts
- Define rotation cycles
- Generate rosters
- Approve claims
- Run payroll
- View reports
- Manage public holidays

Admin Access: Full Create, Read, Update & Delete

3.2 Employee Role

Employees have **limited access**.

Employee Responsibilities:

- View assigned shifts
- Submit claims
- View own payroll
- View claim status

Employee Access: Read Own Data, Submit Claims & No Admin Access

4. CORE SYSTEM MODULES

Developers must build the following modules.

4.1 Authentication Module

Purpose: Secure login and role management.

Required Features: Login, Logout, Password Hashing, Role Validation & Session Management

4.2 Employee Management Module

Used by Admin.

Required Features:

Create Employee, Edit Employee, Deactivate Employee, Assign Employee to Team & View Employee List

Important Notes:

- Each employee belongs to **one team**
- Each employee has **hourly rate**

4.3 Team Management Module

Required Features:

Create Team, Edit Team, Assign Employees & View Teams

4.4 Shift Management Module

Required Features:

Create Shift, Edit Shift, Delete Shift, View Shifts

Required Shift Fields:

Shift Name, Start Time, End Time, Grave Shift Flag

4.5 Rotation Cycle Module

This is a **critical module**.

The system must support **rotation-based scheduling**.

Rotation Example:

DAYS	TEAM	SHIFT
1	Team A	Early Shift
1	Team B	Night Shift
1	Team C	Off
1	Team D	Grave Shift

Rotation repeats automatically.

Required Features:

Define Cycle Length

Assign Shift per Team per Day

Store Rotation Pattern

4.6 Roster Generation Module

Purpose:

Automatically create shift schedules.

Required Workflow:

Select Date Range

Apply Rotation Pattern

Assign Employees

Generate Roster

Store Assignments

Output:

Monthly Employee Shift Assignments

4.7 Claims Module

Used by Employees.

Required Features:

Submit Claim

Edit Pending Claim

View Claim History

Claim Fields:

Claim Date, Shift Type, Hours Worked, Overtime Hours, Public Holiday Flag

4.8 Approval Module

Used by Admin.

Required Features:

View Claims, Approve Claims, Reject Claims & Add Notes

Claim Status:

Pending, Approved & Rejected

4.9 Payroll Module

Purpose:

Calculate employee pay.

Payroll Must Calculate:

Normal Pay, Overtime Pay, Holiday Pay, Grave Shift Allowance & Total Pay

NB: Payroll only includes: Approved Claims

4.10 Compliance Module

Purpose:

Check labour law rules.

Required Checks:

Maximum Daily Hours, Minimum Rest Period, Overtime Limits & Public Holiday Rules

Output:

Compliance Flags

4.11 Public Holiday Module

Required Features:

Add Holiday

Edit Holiday

View Holidays

Holiday data affects:

Payroll calculations & Compliance checks

4.12 Reporting Module

Required Reports:

Employee Report, Roster Report, Claims Report, Payroll Report & Compliance Report

Reports should be exportable.

Formats: PDF/Excel

5. KEY SYSTEM WORKFLOWS

These workflows define **how the system behaves**.

5.1 Employee Setup Workflow

Admin Login → Create Team → Create Employee → Assign Employee to Team

5.2 Roster Generation Workflow

Define Rotation Cycle → Select Date Range → Generate Roster → Store Assignments

5.3 Claims Workflow

Employee Login → View Assigned Shift → Submit Claim → Admin Review → Approve / Reject

5.4 Payroll Workflow

Collect Approved Claims → Calculate Pay → Generate Payroll → Store Payroll Records

6. DATA REQUIRED FROM EXISTING SPREADSHEETS

Developers must ensure the system supports importing existing spreadsheet data.

Required Data:

Employees, Teams, Rotation Cycles, Public Holidays & Claims History

7. TECHNICAL STACK REQUIREMENTS

Developers must use:

Backend:

ASP.NET Core (.NET)/Express (NodeJS)

Frontend:

React

Database:

PostgreSQL

API Style:

REST API

8. SYSTEM PERFORMANCE REQUIREMENTS

The system must:

Load pages within few seconds

Handle multiple users

Prevent data duplication

Maintain fast query performance

9. SECURITY REQUIREMENTS

Must implement:

Password Hashing, Role-Based Access, Secure Login & Protected API Routes

Employees must only see:

Their own data

10. DEVELOPMENT PRIORITY ORDER

Developers must build modules in this order:

1. Authentication
2. Employees
3. Teams
4. Shifts
5. Rotation Cycles
6. Rosters
7. Claims
8. Approvals
9. Payroll
10. Compliance
11. Reporting

11. DEVELOPMENT DELIVERABLES

Developers must produce:

Database Schema, Backend API, Frontend UI, Role Permissions, Payroll Engine & Reports

12. EXPECTED OUTPUT FROM DEVELOPERS

By the end of development, the system must:

- ✓ Generate shift rosters

- ✓ Assign employees automatically
- ✓ Allow claim submissions
- ✓ Calculate payroll
- ✓ Detect compliance issues
- ✓ Produce reports

13. WHAT DEVELOPERS SHOULD DO FIRST

Immediate starting steps:

Step 1 — Create Database

Step 2 — Build Authentication

Step 3 — Build Employee Module

Step 4 — Build Team Module

Step 5 — Build Shift Module